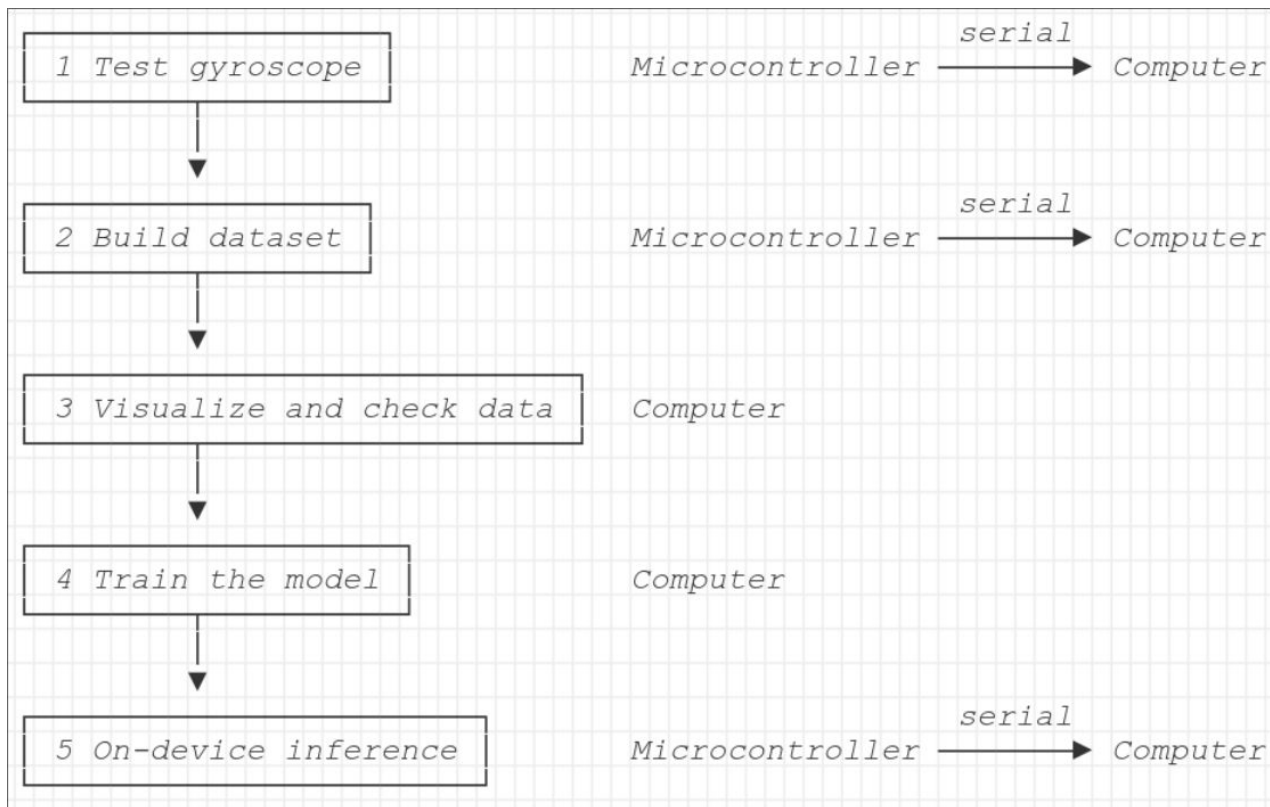


ESP32 gyroscope gesture recognition with edge AI

Made by Marta Viñas and Mateu Magem

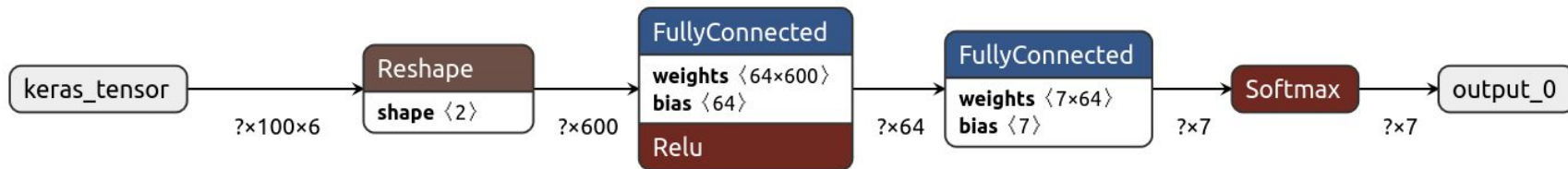
Project workflow and structure



Challenges

- Finding the right architecture
 - Tried CNNs
 - A feedforward neural network was a simpler approach
- Preventing overfitting
 - Gesture simplicity vs. number of gesture data points
- Selecting proper set of movements
 - Added “still” class

Model

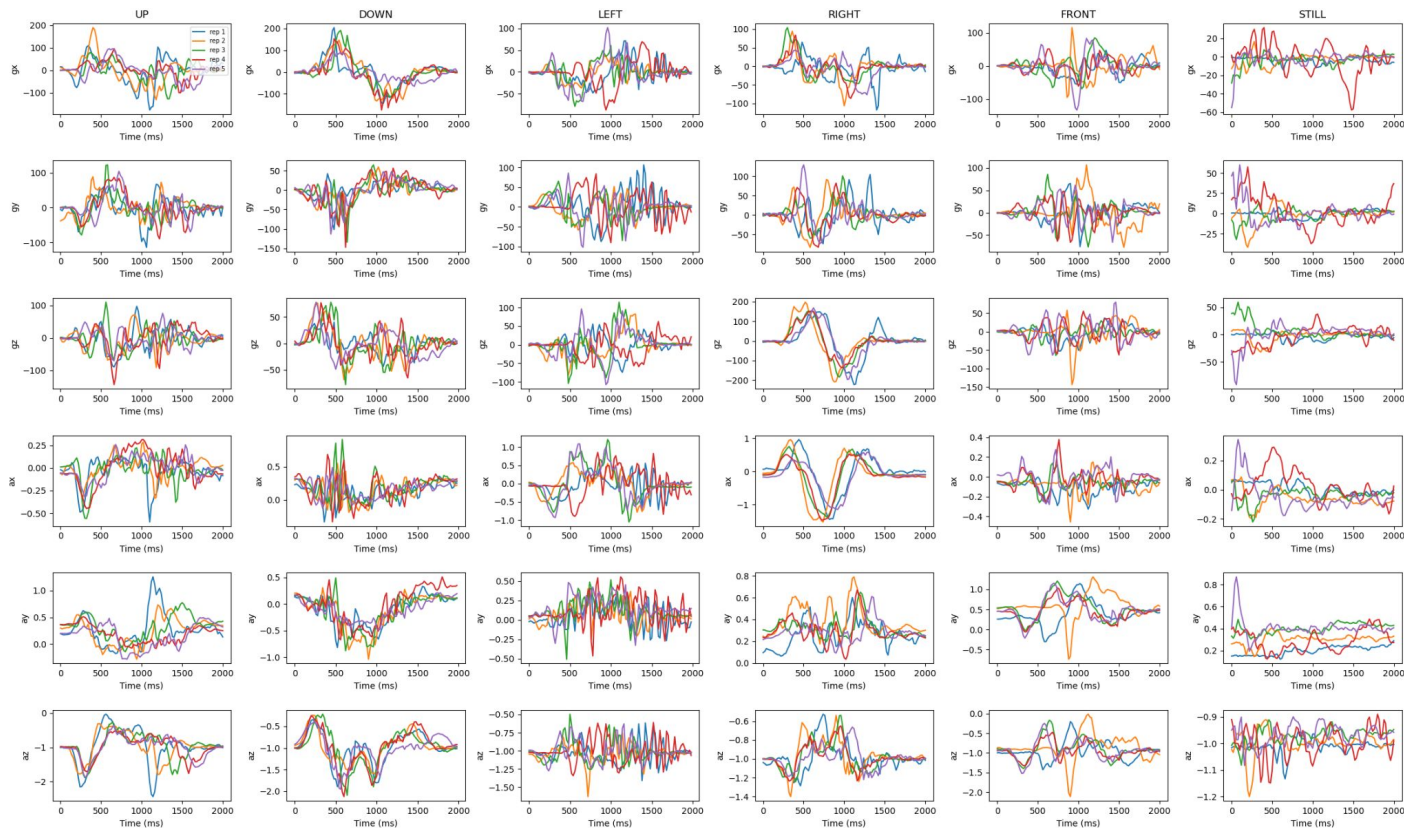


Quantization

- Both *float32* and *int8* models are generated.
- *float32* is used since it is small enough to fit into memory

Data visualization

Gesture comparisons per signal



Demonstration